

A closer look at what/how we can learn from
computational modelling of phonotactics

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May 30, 2025

Abstract

Computational modelling has been extensively used in recent decades in attempts to understand and probe claims about the nature of phonotactic grammars. In this paper, I argue that much more care is needed in inferring aspects of the underlying competence system from model performance, even when compared against human acceptability judgements. I specifically focus on the issue of gradience and show using the Hayes & Wilson (Maximum-entropy based) Phonotactic Model that despite a good fit to acceptability judgements across three prior experimental results focussed on noncewords targetting a variation in word-initial consonant sequences, the model performance does not stem from gradience in the predictions, or gradience in constraints weights, or gradience in the training data (*i.e.*, type frequency information). I further show that a minimal (categorical) phonotactic model does at least as well on all three experimental results. The latter model also reinforces the distinction between the idea of gradience due to the grammar and the grammar itself being gradient. A more general point made is that, as with all theory testing using experiments, it is difficult to learn about the underlying grammatical system from computational models if the models involve auxiliary hypotheses that themselves are not carefully explored and controlled for.

1 Introduction

Native speakers seem to have intuitions about what is or isn't a reasonable nonsense word of their language. As Chomsky and Halle (1968) note, while /blik/ and /bnik/ are both not words of English, English speakers generally prefer /blik/ to /bnik/. There have been many experimental studies that have explored such word acceptability judgments (Albright 2009; Albright and Hayes 2003; Bailey and Hahn 2001; Coetzee and Pater 2008; Coleman and Pierrehumbert 1997; Daland et al. 2011; Freyer 2017; Frisch, Pierrehumbert, and Broe 2004; Gorman 2013; Greenberg and Jenkins 1964; Hayes and Wilson 2008; Ohala and Ohala 1978, 1986; Pierrehumbert 2001, 2003; Pizzo 2015; Scholes 1966; Vitevitch and Luce 1998, 1999; Vitevitch et al. 1997; Wilson and Gallagher 2018, amongst others). The general observation made since the earliest literature on the topic (that I am aware of) has been that acceptability judgements form a cline or gradient, *i.e.*, they are not simple binary intuitions. For example, for English speakers, /blik/ » /bnik/ » /bnzk/ (Chomsky and Halle 1968).

In the last few decades, there has been a huge surge of interest in trying to understand such gradient phonotactic judgments, and many such studies now often include sophisticated computational modelling of the human judgments obtained. Of course, it has long been recognised that linguistic performance, including acceptability judgements, is a multifactorial problem, with grammatical knowledge playing only a partial role in it (Chomsky 1965; Du and Durvasula 2025; Schütze 1996; Valian 1982). For example, Kahng and Durvasula (2023) show that there is a sizeable effect of the gradient in perceptual inferences on acceptability judgements. In a more stark example, Coleman and Pierrehumbert (1997) observed that <mrupation> was rated quite high by their participants despite the nonword having a completely unobserved sequence [mr] that is generally thought to be ungrammatical as it is a member of a class of other #nasal+consonant sequences that are also unobserved — it was likely rated high due to the presence of a very English sounding <-ation> ending. As Hay, Pierrehumbert, and Beckman (2003) point out, participants often decompose input nonwords into morphemes/words. Given the above, gradient in acceptability judgements is inevitable, and it is likely that a sizeable amount of it can possibly be explained in terms of non-grammatical (or performance) factors. Consequently, the presence of (substantial) gradient in acceptability judgements itself doesn't tell us anything about the underlying grammar.

What is at stake with the modern computational models is to see if the actual gradient observed can be predicted. These computational modelling attempts can be classified into a few types: (a) some models presuppose, in contrast to the above discussion, that the observed

gradience in judgements *primarily* comes from within the grammar¹ (Albright 2009; Albright and Hayes 2003; Coetzee and Pater 2008; Coleman and Pierrehumbert 1997; Daland et al. 2011; Freyer 2017; Frisch, Pierrehumbert, and Broe 2004; Hayes 2000; Hayes and Wilson 2008; Mayer *in press*; Pierrehumbert 2001, 2003; Pizzo 2015; Shademan 2006; Wilson and Gallagher 2018); (b) alternatively, other researchers have proposed that observed gradience reflects similarity to existing words, perhaps because the judgment task makes use of a lexical search procedure (Bailey and Hahn 2001; Ohala and Ohala 1978, 1986; Vitevitch and Luce 1998, 1999; Vitevitch et al. 1997); (c) in contrast to the above two modelling strategies, others have shown that computational models with categorical grammars do at least as well (Dai 2025; Durvasula 2020; Gorman 2013; Kostyszyn and Heinz 2021).

The key issues of interest for this paper are to understand what can be interpreted from such computational models and to understand the source of the sometimes good fit of computational models that include an underlyingly gradient phonotactic grammar (meaning, it has gradient constraints). There are a few initial issues that we must address. First, some have argued that there is no phonotactic grammar at all (Reiss 2017), in which case the question of the categoricity/gradience of such a grammar becomes moot. However, given what has been elaborated about this possibility, it is currently empirically too vague to allow for any serious theory comparison. If, in the future, such a position was adjoined with a clear specification of what it is that speakers are doing in such judgement tasks, and if one could draw precise predictions from such a view, then it would be sufficiently useful to study the possibility. Note, I don't mean to question the veracity of the claim of there being no phonotactic grammar at all; it is more that, in the absence of such additional clarifying claims, one must leave the position of there being no phonotactic grammar to the realm of philosophical/logical possibility that cannot be assessed. I return to some possible ways to add substance to this hypothesis in the Conclusion.

The second issue we need to address is that the proliferation of computational models has ironically, in my opinion, made it difficult to see how one makes progress on the issues, and has sometimes muddled the waters. For example, in very recent work, in responding to the success of categorical models of phonotactics, Mayer (*in press*) has suggested that the distinction between gradient and categorical models is a “false dichotomy”. This claim is based on the argument that both can be represented or notated using the formalism of *monoids*. However, the presence of a single formalism that can represent both types

¹In fact, given such models have no non-grammatical factors as modelling parameters at all, one could even say that such models presuppose that *all* the observed gradience in judgements comes purely from the grammar.

of grammars doesn't mean that the distinction is a false one. What is important are the *properties* of the grammars. One can show the error in the claim in a variety of ways. For example, any n -dimensional system can be described with an $n+k$ -dimensional system (where $k>0$, and the n dimensions are the same across the comparison).² This is readily comprehensible by making the additional k -dimensions have a constant value. But, from this, one should not infer that the dimensionality (n vs. $n+k$) is a false dichotomy, since the properties of the n -dimensional system and those of the $n+k$ dimensional system are going to be different.³ Relatedly, the distinction between integers and real numbers is not a false dichotomy just because algebraic variables can represent both of them in the same way. Despite the same variable representation, the properties of the two types of numbers are different, and therefore the distinction between them is real. Closer to the issue of computation is the observation that the distinction between finite state machines and Turing machines is not a false dichotomy just because the latter can be used to represent the languages generated by both the former and the latter. Note, the properties of the two grammars are different. In contrast, there is no distinction between the properties of context-free grammars vs. push-down automata or finite state languages vs. regular languages, even if the notational systems look different (Lewis and Papadimitriou 1998); therefore those distinctions are false dichotomies in a real sense. Returning to the question at hand, the distinction between categorical and gradient grammars is indeed a true dichotomy given that they have different sets of properties (*e.g.*, they have different classes of possible grammars), and therefore it is important to probe the issue both theoretically and empirically.

The third issue we need to address is that there needs to be a clear basis to assess computational models of phonotactics. In this regard, two general strategies have been employed in the literature. One strategy has been to run an acceptability judgement study and fit statistical models with both gradient and categorical factors directly to the judgement data. A general issue with such a strategy is that more complex (superset) models will always fit data better through over-fitting.⁴ This is of course the issue pointed out in footnote 2. We could use a variety of “corrections” (information criteria, prior weighting,*etc.*), but such

²This is identical to saying that the R^2 of a statistical model is non-decreasing, *i.e.*, it cannot decrease, but can increase, with the addition of meaningless terms in the model (Cruise 2019). Therefore, R^2 is not a meaningful measure of model fit by itself — without an evaluation of and correction for model complexity, it is not possible to do model comparison.

³This is of course what makes *Flatland* (Abbott 1884) so entertaining.

⁴As John von Neumann is (potentially apocryphally) said to have claimed: “With four parameters I can fit an elephant, and with five I can make him wiggle his trunk” (see “Von Neumann’s Elephant” 2025, for discussion).

models, at best, provide a compact descriptions of the corpus/lexicon/training data, without a clear pathway to make inferences about actual human knowledge, which is the real question of interest to a linguist, particularly of the generative ilk.

A second strategy that has been adopted in the literature to assess computational models is that of comparing model predictions to human judgments. The basic premise is that it is important to identify the computational generalisations potentially learnt by a speaker from a pre-existing corpus, and then test the predictions of the model against human acceptability judgements. The general procedure for such an assessment is in (1). I will adopt this general strategy in this paper, as it addresses the question of interest related to what human beings know and naturally ameliorates (but not necessarily, removes) issues related to over-fitting raised above.

1. Comparing computational models of phonotactics to human judgements

Step 1: Propose a specific computational model.

Step 2: Train the model on a set of training data or lexicon.

Step 3: Simulate acceptability judgements based on the model.

Step 4: Test model predictions against actual speaker judgements.

The fourth issue is that we need to be clear about what can be inferred when there is in fact a good fit between computational model predictions and human judgements. It has often been assumed that a good fit of model predictions to linguistic data or to human judgements by themselves show us that the theoretically relevant aspects of the model are justified. However, computational models often come with a host of auxiliary assumptions along with claims about the underlying grammar. Therefore, as a matter of logic, one can't automatically assign blame or credit to the crucial theoretical question as opposed to the auxiliary hypotheses for any theoretical comparison, not just computational modelling (Duhem 1954; Lakatos 1970; Quine 1951).⁵ Note, this erroneous argument form must be accepted if one wants to argue that the high performance of a computational model incorporating gradient constraints should be interpreted as evidence that the underlying grammar itself must have gradient knowledge.

⁵This issue is closely related to another famous issue in the philosophy of science, namely, the “tacking paradox” that stems from confirmationist logic being used for scientific inference (Glymour 1980; Mayo 1996) — the paradox that arises is that confirming everything in a model is indistinguishable from confirming just one aspect of the model.

To clarify the logical point, imagine three different computational models represented as functions with multiple arguments/parameters, as in (2). The first two use the same representations, but have a difference in categorical/gradient knowledge (rules or constraints), in rule/constraint definitions, in *how* the multiple rules/constraints interact (additive, multiplicative, *etc.*), and a variety of other parameters. The second and third models are identical except for the fact that latter uses categorical knowledge and the former uses gradient knowledge. In such a scenario, even if Model 2 performs better than Model 1, it is not appropriate to claim it is so because of the gradient knowledge in the model, since there are many parametric differences. It is a rather straight-forward logical error to do so, but one that has been tacitly accepted as a strategy for model comparisons (Dai 2025; Daland et al. 2011; Gorman 2013; Hayes and Wilson 2008, amongst others). In contrast, if Model 2 performs better than Model 3, it is appropriate to claim it is so because of the gradient knowledge in the model, since the only difference between the models is the gradient/categoricity within the grammar.⁶ Essentially, to be able to infer gradient (or any other aspect of the computational model) as the source of the improved fit to human judgements, one needs an *all things kept equal* comparison that is missing in recent model comparisons.

2. Three different models of some linguistic domain

- (a) Model 1: $f(\text{categorical knowledge}, \text{representations}_1, \text{rule/constraint definition}_1, \text{constraint interaction}_1, \text{training set}_1, \text{paramA}_1, \text{paramB}_1, \text{paramC}_1, \dots)$
- (b) Model 3: $f(\text{gradient knowledge}, \text{representations}_1, \text{rule/constraint definition}_2, \text{constraint interaction}_2, \text{training set}_2, \text{paramA}_2, \text{paramB}_2, \text{paramC}_2, \dots)$
- (c) Model 2: $f(\text{categorical knowledge}, \text{representations}_1, \text{rule/constraint definition}_2, \text{constraint interaction}_2, \text{training set}_2, \text{paramA}_2, \text{paramB}_2, \text{paramC}_2, \dots)$

With regard to computational models of phonotactics, one could even pose a starker counter-factual question: what if complex gradient models were *better* than the categorical ones in the literature, should we interpret that as evidence for gradient phonotactic knowledge in speakers? In this paper, I will argue that the right answer to this question is “no”. In fact, as I will argue, we need to do more than this to achieve a deeper understanding

⁶Note, I say “appropriate”, not “correct/true” because the inferred parameter could still be spurious and be just a stand in for another (set of) parameter(s) entirely — but that’s unavoidable with any ampliative inference. However, there is still a path forward. If we repeat the appropriate inference technique with new heretofore unknown parameters, then we will uncover our error, thereby showcasing what Peirce (1931, 1958) called the “self-correcting” nature of science.

of underlying phonotactic grammars when comparing computational models. Specifically, I will argue that researchers have to severely test (Mayo 2018; Peirce 1931, 1958) computational models and in a sense *deconstruct* the models to try and truly understand what is leading to the performance. Simply fitting model predictions against human performance by itself is quite unlikely to further our understanding of these models, and is consequently unlikely to develop well motivated theories of phonotactics, or of phonology/cognition more generally.⁷

The final issue we need to address is that, as mentioned above, there is a panoply of computational models of phonotactics. Furthermore, as pointed out above, one can’t simply compare different computational models that come with a host of other modelling decision-s/parameters. In order to make progress, I will use the Hayes & Wilson (Maximum-entropy based) Phonotactic Model as the computational model reflecting a gradient phonotactic grammar (Hayes and Wilson 2008). I chose this particular computational model because it has generally shown good fits to human acceptability judgements. Moreover, the very fact that the model routinely figures in model comparisons against other computational models of phonotactics is a loud (and nearly unanimous) acknowledgement of the impressiveness of the model and the *de facto* baseline that it has now become. Furthermore, the underlying Maximum Entropy grammar (Goldwater and Johnson 2003) has also been used to model a variety of phonological patterns.

There are five main results I will present: (a) the effectiveness of the original Hayes & Wilson Phonotactic Model and the observation that the model fit improves when the gradient predictions of the model are effectively binarised (Section 3.2), (b) that the gradient in the predictions of the Hayes & Wilson Phonotactic Model doesn’t stem from the constraint weights, as a modified model with no constraint weights does at least as well, if not better (Section 3.3), (c) the gradient in the predictions of the Hayes & Wilson Phonotactic Model doesn’t stem from the lexical type frequency of the relevant consonant sequences in the training data, as a model trained on training data with no frequency structure/differentiation between the consonant sequences does at least as well, if not better (Section 3.4), (d) a nearly degenerate Hayes & Wilson Phonotactic Model with just six constraints and no differentiation in constraint weights does at least as well as the original model (Section 3.5) (e) in addition, in Section 4, I will describe a new model (the **minimal (categorical) phonotactic model**)

⁷As Peirce (1958, Vol. 7, page 231) put it “The process of testing it will consist, not in examining the facts, in order to see how well they accord with the hypothesis, but on the contrary in examining such of the probable consequences of the hypothesis ... which would be very unlikely or surprising in case the hypothesis were not true.”

and then show that its performance is virtually identical to that of all the other models despite being minimally specified. This final result suggests that even if one is committed to the view that some gradience in acceptability judgements is due to the grammar, it doesn't mean that the grammar itself is gradient.

2 Modelling decisions

2.1 Training data

In order to keep as many modelling choices constant as possible and thereby maintain the *all things kept equal* clause of the reasoning, the core training data I will use, following Hayes and Wilson (2008), is a modified version of list of word-initial consonant sequences in the CMU dictionary (Weide 1994). As they note, the list excludes any consonant sequences that they judged to be exotic.⁸ The list of consonant sequences and their type frequencies are listed in Table 1.

Onset	Type Freq.	Onset	Type Freq.	Onset	Type Freq.	Onset	Type Freq.
K	2,764	V	615	Y	268	Z	83
R	2,752	G	537	F R	254	S M	82
D	2,526	JH	524	P L	238	TH R	73
S	2,215	S T	521	B L	213	S K W	69
M	1,965	T R	515	S L	213	T W	55
P	1,881	K R	387	D R	211	S P R	51
B	1,544	SH	379	K W	201	SH R	40
L	1,225	G R	331	S T R	183	S P L	27
F	1,222	CH	329	TH	173	DH	19
HH	1,153	B R	319	S W	153	D W	17
T	1,146	S P	313	G L	131	G W	11
P R	1,046	F L	290	HH W	111	TH W	4
W	780	K L	285	S N	109	S K L	1
N	716	S K	278	S K R	93		

Table 1: Onsets and type frequency used by Hayes and Wilson (2008)

⁸I will return to elaborate on this issue in the Conclusion.

2.2 Test data

I will focus on acceptability judgements of English words where the computational modelling involves the word-initial consonant sequences (“onsets”). One might wonder if it is more appropriate to compare *whole-word* models. I personally am skeptical this is appropriate as whole-words consist of much more representational structure than just segment sequences (if we accept the research in phonology since the 1950s, which I am wont to do). Therefore, unless we are able to model the word in a phonologically sensitive way, there is a distinct danger that conflating/ignoring important representations can lead to spurious fits. Furthermore, there appear to be many other effects on judgements apart from the grammar. For example, as mentioned above, Kahng and Durvasula (2023) found evidence of perceptual gradient affecting judgements. As a second example, Coleman and Pierrehumbert (1997) observed that <mrupation> was rated quite high by their participants despite the nonceword having a completely unobserved sequence [mr], likely due to the presence of a very English sounding <-ation> ending. As pointed out earlier, no current grammatical model predicts this. Therefore, I think it is somewhat inappropriate to test the whole-word predictions of computational models.

Beyond the general issues raised above, the topic of acceptability judgements of English words where the crucial variable is the word-initial consonant sequences forms an extremely focussed problem that does not conflate effects of syllable structure, onset-vowel interactions, *etc.*. Furthermore, this was one of the main foci of Hayes and Wilson (2008), and equally importantly, at this point, there are multiple acceptability judgement studies related to this topic. Specifically, I will use data from (a) Scholes (1966), (b) Albright (2007), (c) Daland et al. (2011). As discussed below, the three different experiments had a variety of different stimuli, experiment designs, and even prompts. A consistent result across all the experiments therefore allows us to, to some extent, sidestep the problem that the inference might otherwise be affected by important considerations related to experiment design, namely the nature of the task (Armstrong, Gleitman, and Gleitman 1983) or the implications of the exact prompts (Firestone and Scholl 2016; Schütze 2005).

The Scholes (1966) data is based on 35 native English speaking seventh graders who responded Yes-No to 66 monosyllabic nonwords varying in consonant sequences, where the rhymes were all bland. The prompt used was “Suppose these are foreign words which English wishes to borrow; which ones will be admitted in their present form and which ones will be changed?” The stimuli were presented auditorily to all students in the classroom at the same time.

The Albright (2007) data is based on native English speaking participants who judged 30 monosyllabic nonwords varying in consonant sequences starting with p-, b- on a 7-step Likert scale, where 1 indicated “Completely impossible as an English word” and 7 indicated “Would make a fine English word”. The rhymes were controlled for neighbourhood density and bigram probability. The task also included 170 filler items (nonce+real). The stimuli were presented auditorily using PsyScope (Cohen et al. 1993) in an in-lab experiment. As an interesting detail, participants were asked to repeat each nonword aloud before rating it — the participant pronunciations were used to exclude any token (for that participant) if the participant didn’t achieve the intended target.

The Daland et al. (2011) data is based on 48 native English speaking participants who judged 96 stress-initial CCVCVC non-words with 48 unique word-initial consonant sequences on a 6-step Likert scale, where 1 indicated “unlikely” and 7 indicated “likely”. The participant were told to “rate them based on how likely it was that the words could become new words of English in the 21st century”. The tail -VCVC sequences were not balanced across stimuli, and consisted of 6 difference sequences [atɪf, ɪbɪd, ɑsɪp, ɛpɪd, ɪgɪf, ɛzɪg] depending on the word-initial consonant sequence. The participants were recruited through Amazon Mechanical Turk, and therefore the experiment was an online (out of lab) experiment. Finally, the stimuli were presented to the participants orthographically.

As mentioned above, following Hayes and Wilson (2008), I modelled only the word-initial consonants for acceptability judgements from each of the three experiments. However, each data set had multiple words with the same word-initial consonant sequences, but with different rhymes. To prevent double counting of the consonant sequences and thereby potentially bias the correlations presented, the participant judgment values for each unique consonant pair were averaged.⁹ These averaged values are presented in Figures 1-3. As can be seen across all three experiments, there is a cline of judgements. And it is the case in all three experiments that the observed word-initial sequences are rated generally higher than the unobserved ones. In the figures in this paper, *grossStatus* indicates whether the sequence is observed (Observed) or not (Unobserved) as a word-initial consonant sequence in the training data mentioned above.

⁹It is worth noting that the ratings in all three experiments for stimuli with identical word-initial consonant sequences were very similar. Therefore, the concern was that any measure of fit would be inflated due to the presence of the repetition.

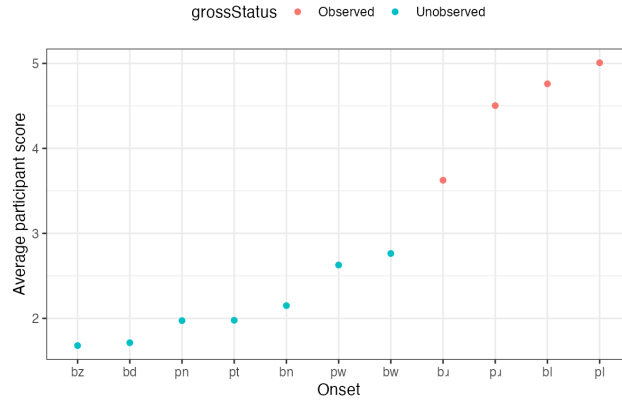


Figure 1: Overall judgements in Albright (2007). Gross status indicates whether the sequence is observed (Observed) or not (Unobserved) in the training data.

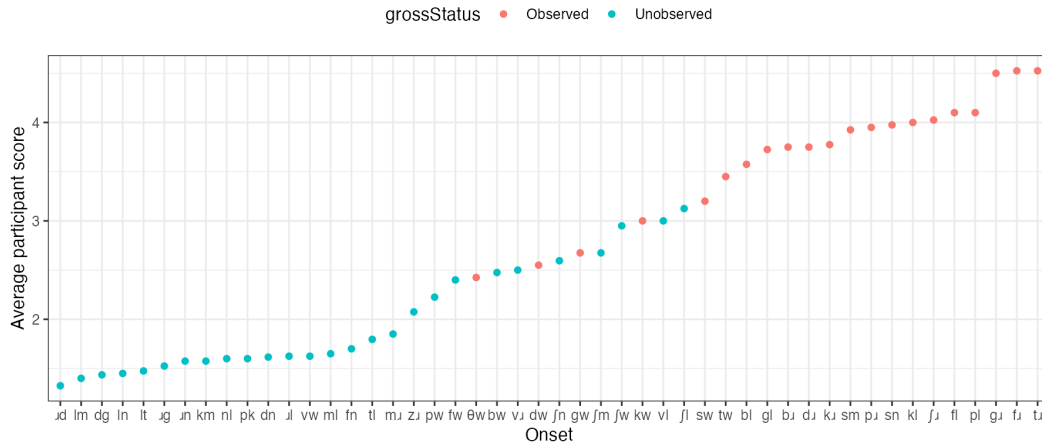


Figure 2: Overall judgements in Daland et al. (2011). Gross status indicates whether the sequence is observed (Observed) or not (Unobserved) in the training data.

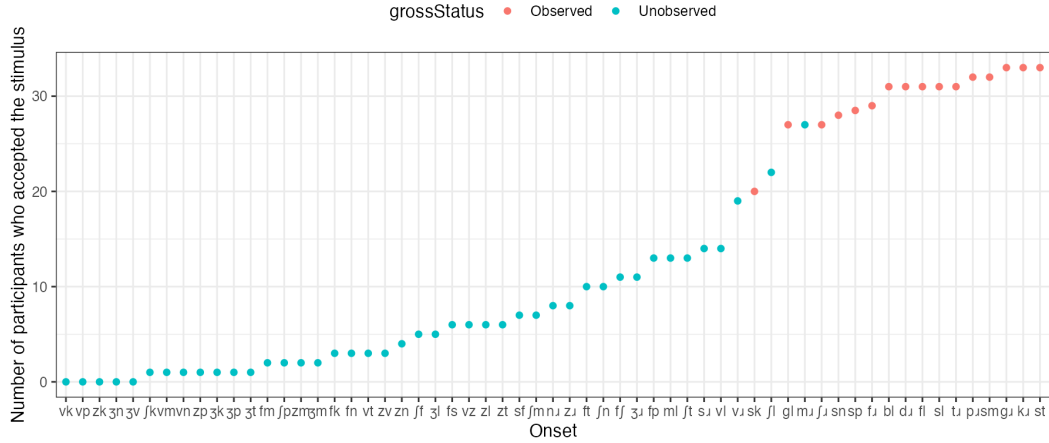


Figure 3: Overall judgements in Scholes (1966). Gross status indicates whether the sequence is observed (Observed) or not (Unobserved) in the training data.

2.3 General remarks about the presentation of results

In this paper, I used the `tidyverse` suite of packages (Wickham 2017) for data analysis and plotting. The statistical analysis was done in base R (R Core Team 2021). Finally, the data and grammars (constraint sets) were converted into $\text{\LaTeX}$ code using the R-package `stargazer` (Hlavac 2018).

Throughout, I will use Pearson’s correlation co-efficient (r) as the measure of fit. While others, such as Dai (2025), Gorman (2013), and Mayer (in press) have used other non-parametric correlation measures, such as Spearman’s correlation or Kendall’s tau, I will maintain the use of Pearson’s correlation co-efficient largely because using non-parametric measures militates against a crucial distinction made in the Hayes & Wilson Phonotactic Model. As explained in Section 3.1, the Hayes & Wilson Phonotactic Model distinguishes between a *harmony* score (a weighted score of constraint violations) and the MaxEnt score or negative exponentiated score (to represent an unscaled value in lieu of a probability value). Using a non-parametric correlation measure will not allow us to distinguish between harmony scores and the negative exponentiated scores, since the negative exponentiation simply reverses the rank-order of the harmony scores.¹⁰ Therefore, to maintain a core theoretical

¹⁰Just to belabour the point with a concrete example. Just imagine a set of values, and another set of the same values that are negative exponentiated. One can easily run the following code in R to see that the Pearson correlations are different for the original and negative exponentiated values, while the Kendall and

distinction of the Hayes & Wilson Phonotactic Model, the Pearson’s correlation coefficient was used as a measure of model fit. Furthermore, as will be seen in the figures presented, the relationships are reasonably linear, so using the Pearson’s correlation coefficient is not inappropriate. However, the Pearson’s correlation coefficient, like a mean, is just a descriptive statistic which doesn’t allow for inferences on its own without associated error probabilities. So, instead of just looking at which model has the highest (Pearson’s) correlation coefficient as has been the case in (all) prior literature, I will consider the 95% confidence intervals for the correlation coefficient in order to see if a specific fit is statistically different from another.

Finally, I will present all the figures presenting the model’s predictions against human performance with the model predictions on the x-axis (therefore, treating it as the independent variable). While this does not make a difference to the measure of fit in this paper (namely, Pearson’s correlation coefficient), it does affect any linear modelling of the relationship. As pointed out by Piñeiro et al. (2008), it is mathematically appropriate to present the predicted values on the x-axis and the observed values on the y-axis, as opposed to the other way around.

3 Understanding the original Hayes & Wilson Phonotactic Model performance results

3.1 Background on the Hayes & Wilson Phonotactic Model

The primary intuitions that the Hayes & Wilson Phonotactic Model is based on are as follows. The model extracts featural ngram generalisations or “constraints” from the actual observed lexicon. Then, it figures out weights to assign to constraints based on the principle of maximum entropy — this is generally thought to be related to the frequency of occurrence of the ngram in the lexicon. The constraints and the corresponding weights are used to get a *harmony score*, which is simply the sum of the products of the constraint violations and

Spearman coefficients are identical for the two types of values:

```
X=rnorm(100); Y = X+rnorm(100); X_negExp=exp(-X); Y_negExp=exp(-Y);
```

```
#Comparing correlations of original values and negative exponentiated values.
```

```
cor(X,Y, method="pearson")-cor(X_negExp,Y_negExp, method="pearson") # ≠0
```

```
cor(X,Y, method="kendall")-cor(X_negExp,Y_negExp, method="kendall") # =0
```

```
cor(X,Y, method="spearman")-cor(X_negExp,Y_negExp, method="spearman") # =0
```

the corresponding constraint weights, as in (3).

3. The harmony score of a candidate

$$h(x) = \sum_{n=1}^N w_n C_n(x)$$

N = number of constraints

$C_n(x)$ = violation score associated with a constraint_n

w_n = weight associated with constraint_n.

Finally, to produce a “judgement” for a nonsense word, the harmony score is then converted into a probability using the formula in (4).

$$4. P(\text{candidate}) = \frac{\exp(-1 * \text{harmony score})}{\sum_{n=1}^{n=\text{all possible words}} \exp(-1 * \text{harmony score}_n)}$$

However, to calculate the denominator in the formula in (4), we need the relevant values of all the possible words in a language — which while theoretically an infinite set, has to be finite for a probability estimation.¹¹ In practice, however, the demoninator is ignored, as it is a constant across all the stimuli, and behaves as a linear scaling factor, so the sum of all scores adds up to one to satisfy one of the axioms of probability. Instead, just the numerator is used as a prediction of the model — this is called the MaxEnt score. I will call the numerator term the *negative exponentiated* score throughout.

Though not often discussed, there are four substantive/theoretical claims in the model, as listed in (5).

5. Substantive claims in the Hayes & Wilson Phonotactic Model

Claim 1: The grammar has gradient knowledge within it (*i.e.*, constraints are weighted).

Claim 2: The effect of the constraints is cumulative (additive in the case the harmony score, and multiplicative in the case of the negative exponentiated scores).

Claim 3: The predicted harmony scores are effectively binarised (a claim that will become clearer later in the paper).

Claim 4: The probability of a particular form is in comparison to all other possible word-forms.

¹¹This step is of course theoretically suspect, and definitely casts a doubt on such models. Furthermore, this is part of a larger general theoretical problem with computational models of phonotactics that see word acceptability as a manifestation of a probability value. I touch on this issue briefly in the Conclusion.

3.2 Results

In this section, I will present the performance of the Hayes & Wilson Phonotatic Model against the three sets of experimental results. The model comparisons will be presented as figures, as all the relevant information is contained in them. For each of the three experimental results, I will present the model’s predictions in terms of harmony scores (*h scores* in the figures), and as negative exponentiated scores (*exp(-h scores)* in the figures). Furthermore, the Pearson’s correlation co-efficient (*r*) and the corresponding 95% confidence interval are presented at the top of the facet corresponding to the relevant score.

As can be seen for all three experimental results, the harmony scores form a cline of values that generally (linearly) correlate with participant judgements, and the correlation values are quite high, suggesting that the fit to human judgements is quite good. Notably, the negative exponentiated scores have a much higher correlation to human judgements for each of the comparisons. Finally, the model does predict that the observed sequences will generally have higher negative exponentiated scores than the unobserved sequences. These results generally replicate the observations of good model fits for the Hayes & Wilson Phonotatic Model in prior work (Daland et al. 2011; Gorman 2013; Hayes and Wilson 2008, amongst others).

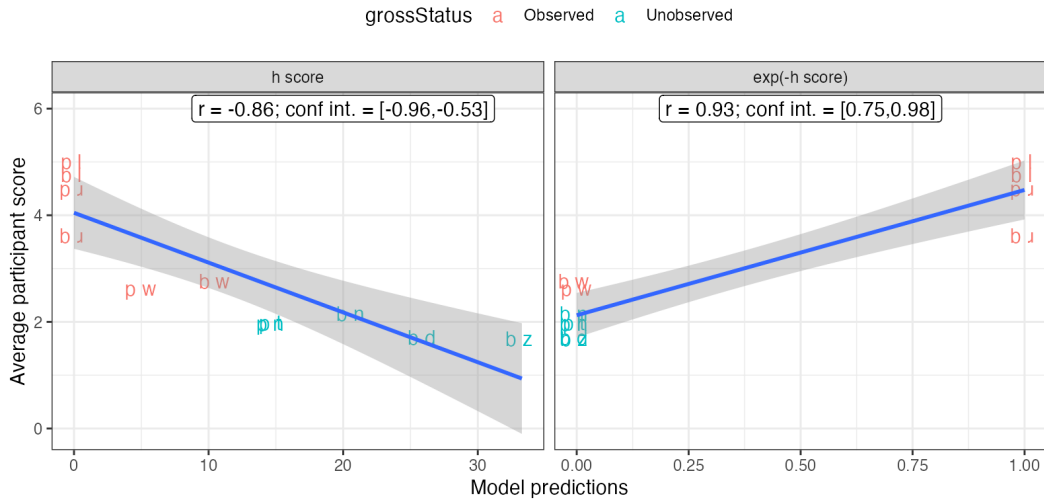


Figure 4: Comparison of the Hayes & Wilson Phonotatic Model predictions to human judgements for the Albright (2007) data.

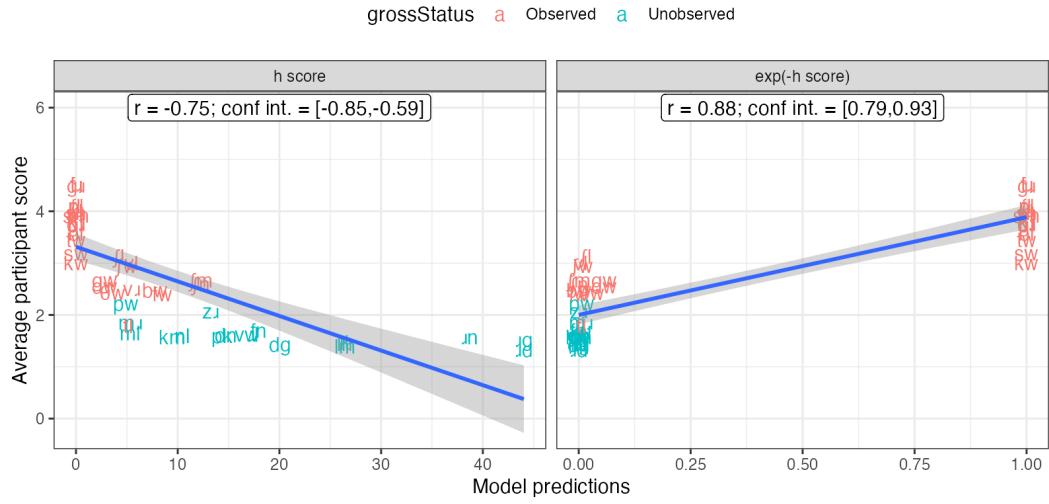


Figure 5: Comparison of the Hayes & Wilson Phonotactic Model predictions to human judgements for the Daland et al. (2011) data.

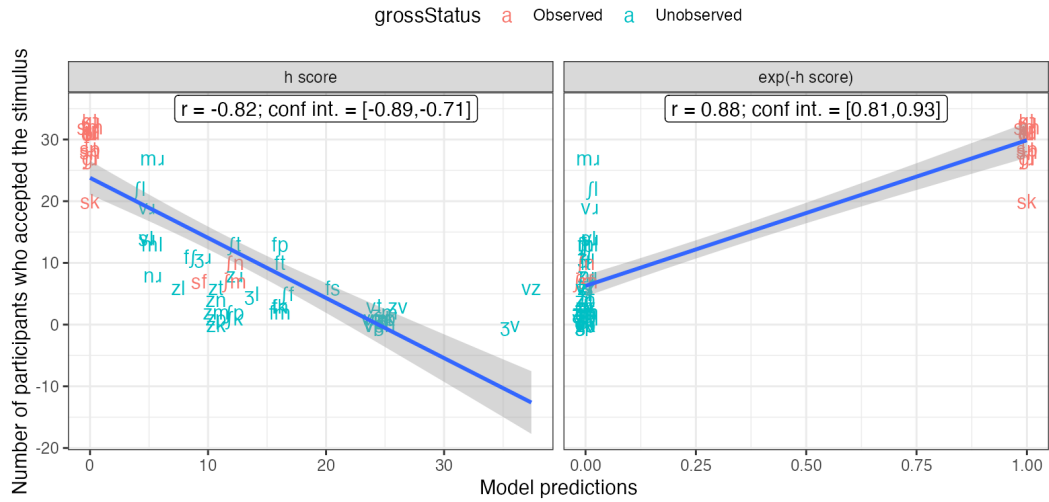


Figure 6: Comparison of the Hayes & Wilson Phonotactic Model predictions to human judgements for the Scholes (1966) data.

The figures, however, make clear that the fits are improving with negative exponentiation largely because the transform results in *less* gradience in the predictions; essentially, the

transform is binarising the predictions — this of course follows straight-forwardly from the properties of the negative exponential transform. This observation is clearest for the Scholes (1966) data where the confidence intervals for the Pearson’s correlation coefficient for the negative exponentiated scores do not trap that for the harmony scores. This effective binarisation of the model predictions observed in the negative exponentiated scores is our first clue that the gradience predicted by the model is not as impressive as what might be thought based on just looking at the model fits. It therefore raises awareness of the need for much more careful model inspection to understand why the model is successful.

3.3 Model check 1: What if the Hayes & Wilson Phonotactic Model had no gradient constraint weights?

The previous sub-section observed that when the gradience is decreased (perhaps, even removed) from the predictions of the grammar, there seems to be an improvement in the fit to human judgements. However, the underlying grammar still possessed gradience in the form of constraint weights. So, one could have argued that the observation in the previous sub-section doesn’t directly speak to the gradient/categorical nature of phonotactic constraints themselves. In this sub-section, we address this concern by simply removing the gradience stemming from the constraint weights (which is the same mathematically as making the constraint weights all the same, and equal to 1 in the original model). So, effectively, the Hayes & Wilson Phonotactic Model now reduces to a modified model where one just sums up the constraint violations for the same grammar (without using the constraint weights). Therefore, the harmony score of a candidate is now modified as in (6), and the subsequent negative exponentiation will be done as before. I will call this modified model the *No-weights* Hayes & Wilson Phonotactic Model.

6. A modified No-weights Hayes & Wilson Phonotactic Model

$$\sum_{n=1}^N C_n(x)$$

In Figure 7, the harmony scores of the No-weights Hayes & Wilson Phonotactic Model are compared against the original harmony scores. As can be seen in the figure, despite the constraint weights differing appreciably, the model predictions of the two models are nearly identical.

3.4 Model check 2: Where do the weights come from and do we need them?

The previous sub-sections showed two things: (a) the fit of the Hayes & Wilson Phonotactic Model was pretty good against human performance; however, the fit seemed to *improve* when the gradient in the *predictions* was *reduced* (or removed even), thereby casting some initial doubt that gradient in the model predictions is truly useful; (b) the gradient in the model constraint weights itself is not useful as a No-weights Hayes & Wilson Phonotactic Model does at least as well as (and, in one case, better than) the original Hayes & Wilson Phonotactic Model.

The two results so far suggest that gradient in the predictions and the gradient in the grammar itself don't improve fits — in fact, if anything, they seem to worsen fits. However, there is still the issue of there being gradient in the training data, in the form of type frequency information about the different word-initial consonant sequences. In this sub-section, we probe this latter issue by removing such frequency information in the training data.

To do this, we first have to figure out what the right frequency-free training data should look like. Table 3 provides a couple of possibilities that should be entertained. In general, we can't directly compare a model trained on a lexicon with type frequency information with a model trained on a lexicon with unique types (see the first and second columns of Table 3, respectively), because that would be an apples-to-oranges comparison, in that there is a massive difference in the training data size. In the example, the training data size for the unique consonant sequence count would be just 3 compared to the 30 for the type frequency count. Instead there's an argument to be made that we need to compare a model trained on a lexicon with equal frequency weighting, as in the last column of the same table, which has the same total size. The main intuition is that we need to deconfound training data size and frequency structure in the training data. It is only the equal type frequency data set that erases frequency structure while retaining the training data size. The same point is made in Sarver (2020). In what follows, I will call the unique consonant sequence count case the *1-frequency* training data; and I will call the latter type the *equal frequency* training data.

Consonant sequence	Actual type freq.	1-frequency	Equal type frequency
k	15	1	10
r	10	1	10
kw	5	1	10
Total	3	30	30

Table 3: A potential training set of unique word-initial consonant sequences

The constraints and corresponding weights for each of the three models are presented in Table 4. The table also provides the constraint weights from a repeated modelling of the Hayes & Wilson Phonotatic Model with exactly the same parameters as the modelling in Section 3. As Hayes and Wilson (2008) note, the constraint weights will differ between different training sessions, even if all the parameters are kept the same, so it is instructive to see model fit and the constraints extracted under repetition to better understand the differences to other types of models.

The first thing to notice about the different models is that the constraint sets are not identical. There is quite some similarity between the original Hayes & Wilson Phonotatic Model modelling (*Original* in the table) from Section 3.1 and the repeated modelling using the same parameters and training data (*Original (rep)* in the table). However, there is some variation in the set of constraints and the constraint weights.

When one compares the original model to the model trained on the equal frequency training data, the constraint sets and constraint weights are more different. Finally, the model trained on the 1-frequency training data is rather different from the original model — it includes just 6 constraints!

Constraint	Original	Original (rep)	EqFreq	1-freq.
*[+nasal,+dorsal]	5.594	5.594	5.370	
*[+sonorant][-word_boundary]	5.257	5.322	6.575	
*[-labial,-strident,-dorsal][+consonantal]	5.148	5.128	5.783	
*[-spread_gl,-anterior,-dorsal][-distributed,-lateral]	4.848	4.927	6.125	1.931
*[+continuant,+voice][-word_boundary]	4.774	2.824	3.725	2.599
*[+continuant,-labiodental,-distributed][+continuant,-round,-lateral]	4.765	5.078	5.413	
*[-continuant,+delayed_release][-word_boundary]	4.735	4.229	4.830	
*[-word_boundary][+delayed_release]	4.469	4.538	5.082	2.068
*[+distributed][+consonantal]	4.177	4.177	4.800	
*[-word_boundary][-nasal,+voice,-round,-distributed,-lateral]	3.577		4.400	
*[+continuant,-spread_gl,-strident][-distributed,-lateral]	3.563	3.569		
*[+voice][-distributed,-lateral]	2.852	2.854		
*[-sonorant,+continuant,+voice,-labial]	2.696			
*[-word_boundary][-sonorant,+voice]	2.580	5.332	3.086	
*[-strident][-round,-distributed,-lateral]	2.361	4.579	2.924	
*[-strident][-approximant]	2.111	1.718	2.425	2.038
*[-sonorant,+continuant,+voice,+distributed]	2.071	4.049		
*[-anterior][-approximant]	1.935	1.705	2.216	1.791
*[-anterior][-round,-distributed,-lateral]	1.277	1.454	1.455	
*[+voice][-approximant]	0.353	0.445	0.755	
*[+continuant,+voice,+strident]		2.688		
*[+delayed_release,+voice][-word_boundary]		2.639	3.296	
*[+continuant,+voice,-anterior,+strident]			4.789	
*[+voice,+coronal][+consonantal]			1.134	
*[+voice][-round,-distributed,-lateral]			0.784	
*[-labial,-strident,-dorsal][-round,-distributed]				1.838

Table 4: Comparing the constraint weights when the Hayes & Wilson Phonotactic Model is trained with actual type frequency data vs. when it is trained with different type of frequency-free training data. A blank cell indicated the model didn't include that constraint.

Original	Equal type frequency	1-frequency
0.88	0.88 [0.81,0.93]	0.88 [0.81,0.93]

Table 5: Correlations of the negative exponentiated scores of different types of modified Hayes & Wilson Phonotatic Models with Scholes’s (1966) data. Numbers in square brackets refer to the 95% confidence intervals.

3.5 Model check 3: Combining 1-frequency training with a No-weights model

As a final model check, in this sub-section, I present the results of a No-weights model that was trained on the 1-frequency training data. Essentially, we are looking at the model predictions of a model trained on 1-frequency training data, namely, the last column of Table 4, where the constraint weights are ignored (so, effectively, equal to 1). Therefore, this is an extreme version of the Hayes & Wilson Phonotatic Model where there is no frequency information in the training data, and there are no model constraint weights either.

As can be seen in Figure 9, the models harmony scores correlate quite well with those of the original Hayes & Wilson Phonotatic Model.

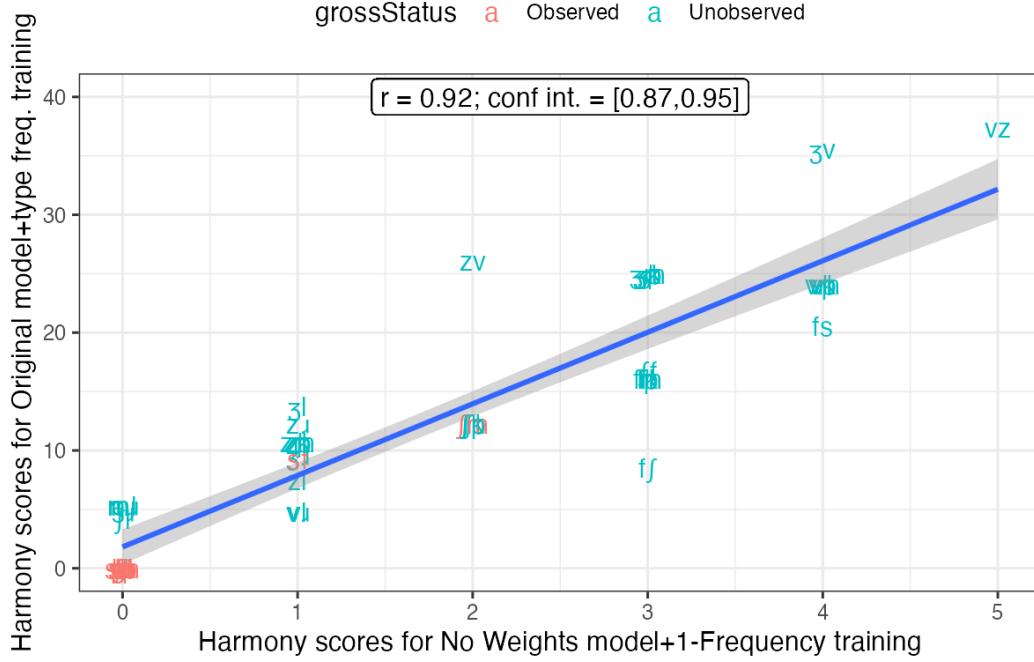


Figure 9: Comparing the predictions when the Hayes & Wilson Phonotatic Model is trained with actual type frequency data vs. a No-weights model trained on 1-frequency training data.

Furthermore, when we look at the model fit against human judgements from Scholes’s (1966) experiment (Table 6), it is effectively the same as that of the original Hayes & Wilson Phonotatic Model.

Original	1-freq. & No-weights
0.88	0.87 [0.8,0.92]

Table 6: Correlations of the negative exponentiated scores of the No-weights Hayes & Wilson Phonotatic Model trained on 1-frequency training data with Scholes’s (1966) data. Numbers in square brackets refer to the 95% confidence intervals.

3.6 Summary of model checks

The model checks gave us three results with respect to the Hayes & Wilson Phonotatic Model: (a) presence of gradience in the predictions worsens fit; (b) presence of gradience

in the grammar (as constraint weights) itself worsen fit; (c) presence of gradient in the training data doesn't affect fit

To summarise, when we look more closely at model checks, we see that it is not clear that the gradient at multiple levels in the Hayes & Wilson Phonotactic Model is really useful. And in fact, there is some evidence that the gradient, at least at some levels, can actually hurt model performance and therefore generalisations themselves.

Crucially, the above model checks maintained the *all things kept equal* clause for model comparisons — therefore, the results can be appropriately inferred over.

4 The minimal (categorical) phonotactic model

The previous section explored the utility (or lack thereof) of gradient at different levels with respect to the Hayes & Wilson Phonotactic Model. However, one should be careful in automatically extending the results to all computational models of gradient. As mentioned in the Introduction, such inferences are generally inappropriate as models come with many other auxiliary modelling/parametric choices that are usually not discussed or explored systematically. For example, the Hayes & Wilson Phonotactic Model has many more built in assumptions about how the constraints are learnt, under what regimes/criteria (Observed/-Expected metrics, simplicity, *etc.*). All of these modelling choices make inference difficult for the theoretically interesting question of the need for gradient within the grammar.

To pursue the question, one really needs to start with a *minimal* model where the auxiliary assumptions are also simple (and preferably strongly motivated, independently). In what follows, I will propose a **minimal (categorical) phonotactic model** that uses features and segments. Such a model gets gradient in acceptabilities without any gradient generalisations. The model is based on a suggestion in Durvasula and Liter (2020). Ideally, in order to perfectly compare the results from the model against the results in the previous section, one should pursue this question within the context of the Hayes & Wilson Phonotactic Model itself. However, the focus here is to ask what is the *minimum* amount of architecture and auxiliary hypotheses necessary. To do this, we are forced to break from using the Hayes & Wilson Phonotactic Model as a framework for comparisons, as it comes baked in with many adjoining auxiliary assumptions that are not manipulable.

As a starting point for the comparison, it is worth thinking carefully about the space of possible grammars (and languages) with the Hayes & Wilson Phonotactic Model learner. With a 100 constraints and unigrams/bigrams/trigrams. This results in a huge number

of possible generalisations, 1-3.5 million in fact. Furthermore, the constraints can also have real-numbered weights. So, the space of possible grammars is technically uncountably infinite. With that as backdrop, it is worth asking what is the minimal model we can get away with in predicting human performance, and can we do it with minimal auxiliary assumptions (and preferably strongly motivated independently)?

The minimal (categorical) phonotactic model is a phonotactic model that contains all the observed single feature unigrams/bigrams and segment unigrams/bigrams in the learning corpus.

7. Example generalisations in the minimal (categorical) phonotactic grammar.

- *e.g.*, p r
- feature unigrams: [+labial], [+voiceless], [+son], ...
- feature bigrams: [+labial] - [+son], [+voiceless] - [+liquid], ...
- segment unigrams: p, r
- segment bigrams: #p, pr

Note: this is the minimal set of things needed by any grammatical model that has access to features and segments.

Within such a phonotactic model, any complex feature unigrams: $\begin{bmatrix} +labial \\ +voiceless \end{bmatrix}$ or complex feature bigrams: $\begin{bmatrix} +labial \\ +voiceless \end{bmatrix} \begin{bmatrix} +alveolar \\ +voiced \end{bmatrix}$, *etc.* are simply not learnable. So, the space of possible languages is quite small. With 25 consonants (including word boundary as a symbol) and 21 binary features, the unigram+bigram search space = 25 segments unigrams + 26*26 segment bigrams (including word boundaries) + 21*2 binary feature unigrams + (21*2+1)*(21*2+1) binary feature bigrams (including word boundaries) = 2592 generalisations.¹² Furthermore, there are no weights to estimate. So, this is a *finite* space of possibilities. An important consequence of such a model is that it immediately gives us a three-way distinction between *observed and grammatical* » *accidental gaps* » *ungrammatical*.¹³ I label the model the minimal (categorical) phonotactic model because it is the smallest model that achieves the desideratum of the grammatical knowledge itself predicting

¹²With trigrams: it increases to 99,675 generalisations

¹³Note, the model does make the claim that all observed sequences are grammatical. However, note that some “exotic” sequences were removed from the training data. I return to this important issue in the Conclusion.

this trichotomy. With accidental gaps, while the featural combinations might all be present, the actual segment sequence will still be missing from the phonotactic grammar.

Next, we need (preferably) independently motivated auxiliary assumptions to get the grammar to make contact with acceptability judgements. Essentially, we need a linking hypothesis about *performance*. The performance related linking hypothesis is as follows: for each nonsense test word, one counts the number of featural and segment unigrams/bigrams that are not in the grammar, and then applies add-1 Laplace correction (to avoid the problem of zero counts).¹⁴ Finally, a $\log(\text{corrected count})$ is calculated. This last step is of course a very well-motivated general principle in cognition, namely the *Weber-Fechner law* (Sun et al. 2012). There are no additional (or free) parameters to fit to the data, and therefore there is no wiggle room in the predictions.

4.1 Results

In this sub-section, I compare the predictions of the minimal (categorical) phonotactic model with the acceptability judgements from all three sets of experimental results under consideration in this paper, and with the model fits of the original Hayes & Wilson Phonotactic Model that were presented in the previous section. When comparing the model’s results to the Hayes & Wilson Phonotactic Model, I will compare both the model scores themselves and the negative exponentiated version of the scores, so that the comparison is maximally similar to the modelling choices of the Hayes & Wilson Phonotactic Model. Again, the attempt is to, as much as possible, compare apples to apples, or more generally, a member of fruit-class_x to another member of the same class.

As can be seen in Figures 10-12, the minimal (categorical) phonotactic model fits the human judgements quite well.

¹⁴Note, this correction is not independently psychologically motivated, and therefore it can and should be debated. But, I will for current purposes go along with it, since some sort of 0 frequency correction is standard in computational modelling. We can however weakly motivate this choice as piggy-backing on the Weber-Fechner law. If, indeed, some sort of logarithmic transform is done over counts, then one needs non-zero counts, as $\log(\text{zero})$ is not defined. And since counts are integer-valued, the smallest addition one can imagine is 1 to solve the problem — this motivates the 1-Laplace correction.

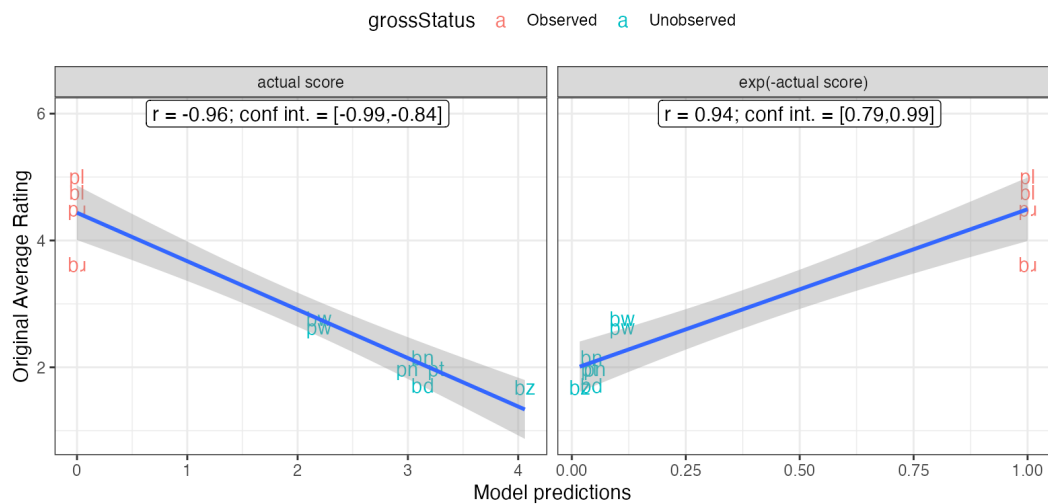


Figure 10: Comparison of the minimal (categorical) phonotactic model predictions to human judgements for the Albright (2007) data.

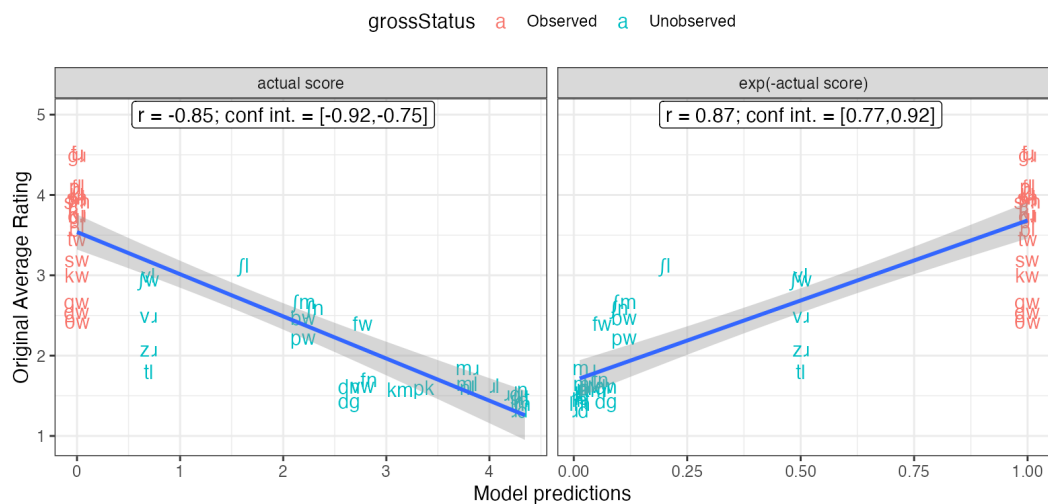


Figure 11: Comparison of the minimal (categorical) phonotactic model predictions to human judgements for the Daland et al. (2011) data.

the Hayes & Wilson Phonotactic Model which has an incredibly rich architecture.

5 Conclusion

In this paper, I presented modelling results and model checks related to the Hayes & Wilson Phonotactic Model and modelling related to a new **minimal (categorical) phonotactic model**. With respect to the Hayes & Wilson Phonotactic Model, though it has been touted as evidence of gradient phonotactic knowledge, it was observed in a series of model checks that claims of evidence of gradience are not sustainable, at least for the case of English word-initial consonant sequences — the very case that was used to show evidence for the model in the first place. There was little to no evidence that gradience in the predictions or in the constraint weights or in the training data appreciably improved the model fit to human judgements. There was in fact some evidence that gradience of the predictions and/or constraint weight hurt model fits to human performance. The results obtained are reminiscent of the results in Dai (2025), Gorman (2013), and Kostyszyn and Heinz (2021), who all argued that the well-specified categorical models do just as well as gradient models. However, the focus in the current paper was to show that the touted performance of the gradient model (in our case the Hayes & Wilson Phonotactic Model) itself doesn’t come from any gradience in the predictions or grammar or training data.

In previous work arguing for categorical models, the gradience observed in the judgements was largely chalked up to (although, not discussed elaborately) noise. A useful extension of this idea is to interpret “noise” as a cover term for various unidentified non-grammatical sources. In fact, one such non-grammatical source is the influence of perception in judgements. As Kahng and Durvasula (2023) show, perceptual gradience has a sizeable effect on the ratings of nonwords, and therefore, should be factored in in interpreting gradience in acceptability judgements. However, even if one were to entertain the possibility that some of the gradience in the acceptability judgements stems from the gradience in the predictions of the grammar, it doesn’t follow that the grammar itself has *gradient knowledge* (or constraints in our case). To make this argument clear, I proposed a minimal (categorical) phonotactic model with very few auxiliary assumptions and no free parameters at all. This latter model fit the human judgements at least as well as the more complex Hayes & Wilson Phonotactic Model. Furthermore, the model predicts some amount of gradience in judgements along with getting the important distinction between observed and grammatical » accidental » ungrammatical. In short, a grammar can lead to systematic gradient accept-

ability judgements, yet the knowledge (or constraints) within the grammar themselves are categorical.

Categorical models can be thought of as strict subsets of gradient models (the constraint sets are the same, but lack any information about gradience). Given that they form strict subsets, they form smaller hypothesis spaces. In fact, given the lack of gradience, the hypothesis space is really speaking finite, compared to the uncountably infinite space of gradient models. So, gradient models generally have much more *flexibility* than categorical models in accounting for data. However, it is important to separate accounting for data from *explaining* the data. As pointed out in the Introduction, a more complex (superset) model will always account for more data simply because it has more flexible parameters, but what matters is how well it explains the data or for that matter truly predicts unseen/untrained data (*i.e.*, when severely tested). Therefore, while flexibility can be seen as a virtue in some domains, it is distinctly not a virtue in theory construction where we want to *understand* the system. Furthermore, as has been extensively argued in the philosophy of science literature, simplicity is a virtue of a theory (Goodman 1943, 1967).¹⁵ Goodman further argues that simplicity is an important criterion one employs in distinguishing between achieving a true scientific statement and simple curve fitting. While, no doubt, one can easily arm this discussion with an opposing list of other philosophers who disagree with Goodman’s claims, there is a core aspect of the claim that I believe is worth retaining, particularly with respect to the nature of gradience in phonotactics or phonology, or linguistics more generally. Say, it is in fact the case that gradient models do much better than categorical models with another sub-domain of phonotactics, one should not see the flexibility of gradient models as a benefit. If anything, it is the fact that we have entertained categorical models that forces us to ask the important question of *why* the sub-domains have such different results, and in doing so will open up new avenues of research and new research hypotheses pushing the field forward. Therefore, entertaining simpler, more parsimonious, models allows us to explore the infinite space of scientific theories in a tractable fashion. An alternative research strategy, to be perfectly consistent with seeing flexibility as a benefit, is one where one chooses the most expansive model with the argument that it is most flexible; however, such a model is ill-defined given the space of possibilities.

The results in the paper suggest that likely much more can be learnt from the nature of the constraints/rules and the representations used in the grammar, than the issue of gradience itself. That is, the results to my eye suggest a return to traditional phonological

¹⁵“Simplicity, in at least some respect, is a test of truth.” (Goodman 1963).

concerns, largely because they are likely to bear (bigger) fruit. For example, the minimal (categorical) phonotactic model allows for only single feature bigrams (or n-grams) and segment bi-grams — this is a rather limited constraint space. Note, the model was not only found to be predictive in this paper, but it was also found to be useful in understanding the artificial language results in Durvasula and Liter (2020), where they observed that participants are able to learn multiple single-feature bigram generalisations, but not more complex multi-feature bigram generalisations. However, it is worth pursuing a project where the constraint space is systematically modified, while keeping everything else constant, to see if there is an improvement in model performance when compared against human judgements. Given how well the minimal (categorical) phonotactic model actually performs in this regards, perhaps the improvement will be incremental, in which case it is important to ask why the increase in constraint/rule space leads to only incremental improvements and if we should be deriving the increased model fit through other non-grammatical factors.

In line with the general advice in this paper that we should be careful in inferring aspects of the grammar from computational models, one can turn a critical eye even towards the minimal (categorical) phonotactic model that I proposed, and try to decompose the effects into an even simpler model, where there are only single feature unigrams/bigrams in the phonotactic grammar, and segmental generalisation fall out of extra-grammatical systems. Note, the juxtaposition of both segmental and featural representations within a grammatical model is quite weird if segments are seen as epiphenomenal constructs based on features. Therefore, we should be probing this issue further. Perhaps, the segmental knowledge is actually from orthography/orthotactic knowledge and not phonological knowledge *per se*. There is by now a fair bit of evidence that there are orthographic effects in speech production, perception and even in loanword patterns (Grippando 2019; Stoehr and Martin 2022; Vendelin and Peperkamp 2006; Warner et al. 2004) — all showing that orthographic and orthotactic knowledge affect performance, which is thought to be multi-factorial anyway. So, though currently just a speculation, it wouldn't be surprising at all if orthotactic knowledge is the ultimate source for the segmental effects in acceptability judgements. While fairly specific orthotactic models are required to make progress, it does highlight the main point of this paper that a good computational model fit should not be directly interpreted as evidence of specific theoretical constructs (included within the computational model) without proper model checking/inspection and understanding non-grammatical factors, particularly when it means we are enlarging the space of possibilities for the grammar.

Furthermore, here is potentially where the no-phonotactic grammar view that Reiss

(2017) espouses does have a prediction to make. If that view is right, and if segmental (not featural) knowledge is what is driving judgements, and is itself from orthotactic knowledge (and not true phonological knowledge, *per se*), then we should see a difference between literate and illiterate speakers. While such tests are quite minimal in modern linguistic research, there is at least somebody of work discussing similar questions related to literacy (Linell 1979), but see Kazanina, Bowers, and Idsardi (2017) for an alternative view of how orthography might only indirectly affect performance by essentially making speakers more meta-linguistically aware. The bottomline is that if we are interested in learning about the underlying competence/grammatical system, we need to be as critical as possible, and only allow new/more constructs into the system if they are theoretically unavoidable and can be justified through repeated experimentation and analysis — the simple fact of just accounting for the data is far from sufficient to make the case. Furthermore, theorists and computational researchers have to take performance factors seriously and incorporate such effects in the modelling or devise experiments that somehow sidestep such concerns or come up with the relevant *gedanken* experiments that go beyond the concerns to make an appropriate inference. While others might disagree, this has been the model for the rest of the natural sciences from my lens, and therefore, I see benefit in emulating that approach - not because there is a guarantee, but because there is no better way that I am aware of. Either way, just acknowledging non-grammatical factors and then simply ignoring such factors and fitting computational models instantiating specific grammatical views seems somewhat inappropriate¹⁶

Finally, and related to the last point, many computational models, incorporating either gradient and categorical grammars, include purely segmental representations (Dai 2025; Gorman 2013; Mayer *in press*, amongst other). It is worth noting, in line with the previous point that this is a remarkable departure from modern phonological representations, which are almost completely feature- or segment- or element-based (Chomsky and Halle 1968; Goldsmith 1976; Kaye, Lowenstamm, and Vergnaud 1985). For the same reason mentioned in the Introduction, related to the difficult of making inferences about the underlying grammar when other decisions are different, it is difficult to know how much of the modelling success crucially depends on the representations chosen; therefore, it is difficult to interpret such computational models as saying anything about phonological grammars as conceived in current day terms. As with the previous point, it would be optimal to derive the seg-

¹⁶There is a distinct fear that simply ignoring non-grammatical factors will not allow us to make progress and the multitude of computational models will just increase without resolution. As a result, we as a field are stuck between playing whack-a-mole and chasing our own tails — both entertaining, neither productive.

mental effects ultimately from non-grammatical sources (including orthotactics), if one is committed to features or autosegments or elements as the true phonological primitives. Alternatively, one can discard those representations in favour of segmental representations as the true phonological primitives.

5.1 Additional issues, clarifications and speculations

There are five more related issues worth elaborating on. First, sometimes computational models come with *free parameters*, parameters that are simply fit to the very same data that the models are trying to predict. For example, the Hayes & Wilson Phonotactic Model comparisons have often been accompanied with a separate *temperature* parameter, which is a non-linear scaling parameter used on the predictions that allows a better fit to human judgement data. However, the use of such free parameters actually makes it harder to understand which aspects of the model are doing the grunt work. In contrast, the minimal (categorical) phonotactic model proposed is not meant to be used with any free parameters, and in that sense, allows for a more severe test of the theory.

Second, the results in this paper should not be seen as a falsification of phonotactic grammars incorporating gradience, but as a foray into *understanding* where the gradience in acceptability judgements comes from. Future work might very well reveal that there are other domains of phonotactic judgements where gradient grammars fit the judgement data better. However, in line with the note of caution throughout, this should not be interpreted as evidence of gradient phonotactic grammars without extensive model checking and replication. Furthermore, if some domains show that gradient phonotactic models fit the judgement data better, and other domains show that categorical models are better, this is chance to understand *why* that is the case.

For example, recently, Mayer ([in press](#)) presented results from a very careful study that compared model performance against native Turkish speakers' judgements of CVCVC non-words. Specifically, they compared models incorporating bigram information over vowel sequences (strict locality was not necessary for the calculation). The gradient model incorporated bigram frequency information, while the categorical models only had access to the presence/absence of the bigram. They observed that the gradient model fit the human judgements better. There are three issues that we need to put aside even though they are extremely relevant: (a) that probabilistic grammars that model acceptability as a product of bigram probabilities make a deeply problematic prediction that longer words will be more ungrammatical (see Alves [2023](#); Coleman and Pierrehumbert [1997](#), for discussion); (b) that

it is not clear that the gradient bigram model that Mayer (in press) used actually fit the human judgements better than the other models they considered, since confidence intervals for the fit measures were not provided; (c) that the gradient bi-gram model was not model checked (as was done in the current paper) to see what the source of the better fit was. I put the above issues aside though they are quite relevant, because even if one grants the above are not problems, there is still an experimental issue to consider. Given the results in the current article and the fact that Daland et al. (2011) found an identical bigram model did worse than the Hayes & Wilson Phonotactic Model for their English results¹⁷, how does one square the results from the Turkish judgments? Note, the stimuli in Mayer (in press) involve CVCVC words, where almost all the vowel sequences are observed in the training corpus. Table 8 shows the observed V-V frequencies in the Turkish Electronic Living Lexicon (Inkelas et al. 2000).¹⁸ This is in stark contrast to the stimuli in Albright (2007), Daland et al. (2011), and Scholes (1966), where a fair portion of the nonwords contained completely *unobserved* sequences. So, the task in Mayer’s (in press) experiment is substantially different from the one participants performed in the experiments used in the current paper. The former involved almost completely observed segment sequences, while the latter didn’t.¹⁹ We know that in other domains, participants often re-interpret tasks and provide inappropriate evidence of specific claims (Armstrong, Gleitman, and Gleitman 1983; Firestone and Scholl 2016; Schütze 2005). The participant in Mayer’s (in press) experiment is perhaps forced to re-interpret the task as deciding between which observed sequence is a *better* word of Turkish — when the grammar is silent on the distinction, the participant is forced to depend on lexical similarity effects, perhaps. If so, it is not surprising that there is more of an influence of gradient information in Mayer’s (in press) results. Note, this is a possibility that deserves further empirical probing. For now, I just raise it to point out that the real challenge is in figuring out how to interpret conflicting fits to human judgements from different phonotactic domains/questions.

¹⁷Albeit, it is not clear from their results that the Hayes & Wilson Phonotactic Model fit was statistically significantly better than that of the bigram model.

¹⁸I extracted these frequency counts from the file labelled `MASTER.db.txt` in the online database available at <https://linguistics.berkeley.edu/TELL/>.

¹⁹This distinction is akin to asking a participant to judge sentences as acceptable when all the sentence frames in the experiment are observed (and thus likely grammatical), versus asking them to judge observed vs. (likely) ungrammatical sentences. The tasks are quite substantially different at face value.

	a	e	i	o	u	y	ø	ʊ
a	4,186	1,023	1,463	387	648	113	52	2,014
e	1,424	2,024	2,036	194	258	104	27	44
i	1,035	780	1,165	219	39	48	21	50
o	981	183	163	182	557	29	25	11
u	1,120	165	103	32	698	20	12	18
y	418	1,063	119	24	92	589	18	14
ø	49	442	14	25	8	256	17	8
ʊ	583	63	51	18	8	6	10	659

Table 8: V₁-V₂ frequencies in the TELL corpus

Third, in this paper, to keep as many aspects of the model comparison identical as possible, I maintained Hayes and Wilson’s (2008) original modelling decision of limiting the training word-initial consonant sequences to a subset of what is actually observed in the CMU dictionary. As mentioned in Section 2.1, Hayes and Wilson’s (2008) rationale was that they wanted to remove any consonant sequences that they judged to be “exotic”. It is for this reason that both the Hayes & Wilson Phonotactic Model and the minimal (categorical) phonotactic model probed in this paper are not learning models *per se*, but grammatical models. From a learner’s perspective, the relevant question is of course, one of identifying the exotic consonant sequences in the first place. Here, one could make an argument that some sort of lexical frequency information is valuable to identify the set of sequences to learn from. This is exactly the tack taken by Dai (2025).²⁰ However, the issue of using some frequency information (if at all) to identify the sequences to generalise over should be kept separate from the issue of the nature of the grammar itself that is learned from such a set of inputs or the issue of whether frequency information is used once such a set of sequences has been identified. Therefore, it is possible that frequency information is used for subsetting to the relevant set of stimuli to learn over, but not for the sake of utilising the observed frequencies to learn the grammar *per se*. Even here, there is a second alternative — one needs to look at the words such “exotic” consonant sequences appear in. If they appear in words that are unlikely to be experienced by a learner within a certain critical (age) window (Lenneberg 1967), then perhaps the learner fails to include them in the set of sequences to learn over. For example, take the case of the word-initial consonant

²⁰This is an approach that has been espoused in other work on the productivity of linguistic generalisations (Yang 2016).

sequence [sf], the CMU dictionary has a total of 12 words with that sequence.²¹ I suspect the likelihood of an infant being exposed to any of those words is extremely low. If so, then there is no need for frequency information even in identifying the relevant set to learn over. For this possibility, one needs to look at the actual input that children get within the first few years of their lives.

Fourth, some of the aspects I touched on in this paper are awfully similar to the *bounded rationality* work by Gerd Gigerenzer and colleagues (Mousavi and Gigerenzer 2014), where they use the distinction between *risk* and *uncertainty* (Knight 1921) — *risk* refers to the case where the full sample space and associated probabilities are known or fully estimable, as in games of chance, while *uncertainty* refers to more real world conditions where the sample space is not fully specified, or therefore probabilities are difficult (and perhaps impossible in a technical sense) to estimate. Gerd Gigerenzer and colleagues have argued that complex/gradient optimisation perform better in the former case of *risk*. However, simplicity and heuristic thinking perform better in the latter case of *uncertainty*, where estimation is less accurate, particularly when the predictive uncertainty of the problem and the number of possibilities is large, but the size of the learning sample is small (Gigerenzer 2008). The case of phonotactics in natural language clearly falls in the *uncertainty* camp, and therefore it is quite heartening to see the convergence between the results in the current paper and those of general cognition related to decision making.

Finally, the main point I have made in this paper also applies to recent comparisons of large language models such as ChatGPT to human performance (Piantadosi 2023; Qiu, Duan, and Cai 2025). It has been argued that such models do not have explanatory value as they don't truly *explain* human language patterns, but merely reproduce them as they are in the training data; and if the training data were contrary to systematic constraints of human language, then the performance of such models may very well still mirror that training data (see Chomsky, Roberts, and Watumull 2023; Hunter 2025; Kallini et al. 2024; Kodner, Payne, and Heinz 2023, for elaborate discussion of this issue). The results in this paper add to that criticism. Even if such models addressed the above problem, they still are quite far from desirable as *theories of language* (or syntax, more precisely) because we are unable to decompose such models to see which aspects of the models are driving the results. Therefore, there is a genuine problem with success-attribution in such models.

²¹The full list is as follows: <sfernice, sferra, sferrazza, sfed, sforza, sfuzzi, sphar, sphere, spheres, spherical, spheroid, sphinx>. Many of these words are unknown even to me, let alone an infant.

6 Acknowledgements

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